Arem Praveen Prakash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.10
RGUKT IIIT RK Valley	Class XII (Central Board of Secondary Education)	2019	8.17%
Montessori High School	Class X (Telangana Board of Secondary Education)	2017	9.80%

ACADEMIC RESEARCH

Molecular Dynamics (MD) Simulations of Elastin-Like Polypeptide (ELP) Drug Delivery Systems | Master's Thesis |

Dept of Biotechnology, GoI funded project | *Guide: Prof. Saptarshi Majumdar* |

- Developed **20+** MD simulations pipeline across 280-350K to investigate **drug behavior**, structural stability near tumor cells in human body
- Reduced **70%** of simulation setup time by building a Bash script for end-to-end MD simulations, enabling rapid execution at conditions
- Processed **50+** time-series datasets using **Python, NumPy, and Pandas** for statistical analysis, pattern identification & molecular hypothesis
- Built a **predictive ML model** by integrating AI/ML + MD simulations, leveraging 50+ time-series datasets to predict protein characteristics

TECHNICAL SKILLS

Languages, Libraries & Tools: ***Python, Excel, SQL****, Java, MATLAB, Bash/ Shell*

Data Science: ***Machine Learning****,* ***Neural Networks****,* ***GenAI, LLMs****, NLP, LangChain, PyTorch, Scikit-learn, RAG*

Relevant Courses: *Optimization, Advanced Numerical Analysis, Introduction to Data Analysis, Calculus, Linear Algebra*

Certifications: *Data Science Bootcamp 2026 (365 Careers - Udemy), Information Technology (8 months - IIIT RK Valley)*

TECHNICAL PROJECTS

Machine Learning for Sustainable Concrete Design | Optimization |

Academic Project | *Guide: Prof. Kishalay Mitra* |

- Developed a **3-layer ANN model** on **1,000** concrete samples using **ReLU** (Rectified Linear Unit) activation to optimize strength and energy
- Configured **200** epochs of ANN training with a batch size of **16** using **Adam optimizer**, enabling stable **weight updates** for prediction
- Optimized **100** concrete formulations using **NSGA-II** (Non-Dominant Sorting Genetic Algorithm) **~200** generations for suitable candidates

Molecular Descriptor Analysis for Yield Prediction | Chemical Processes |

Academic Project | *Guide: Prof. Sharad Bhartiya* |

- Built **4** ML models, with **XGBRegressor, Neural Networks, SVM & Random Forest**, to predict reaction yield from 1,600+ compounds

- Applied **K-Means clustering** to molecular descriptors for grouping compounds into distinct clusters based on molecular-properties similarity
- Achieved an **R^2 0.87,** comparing multiple ML models for reaction-yield prediction, identifying XGBRegressor as best-performing model

Toxicity prediction using Machine Learning | *Predictioneer* |

Top 5 out of 50+ participants | *Azeotropy IIT Bombay* |

- Developed an **ML framework** leveraging **8,459** chemical compounds for effective & robust prediction of compound's LC50 toxicity levels
- Applied **Principal Component Analysis** (PCA) to reduce **33** molecular descriptors to **23** key components while retaining **~90%** of variance
- Addressed class imbalance in toxicity data using Synthetic Minority Oversampling Technique(**SMOTE**), improving class representations

POSITION OF RESPONSIBILITIES

Placement Coordinator | Chemical Engineering |

Office of Career Service (OCS) Member | *Placement Cell* |

- Coordinated and pitched to **20+** international companies, including Japanese organizations, to drive campus recruitment & collaborations
- Developed **10+** customized pitch decks for recruitment outreach, aligning student talent pool with company-specific hiring requirements

Teaching Assistant | Department of Chemical Engineering |

Courses: Transport Phenomena, Thermodynamics, Aspen Simulations Lab |

- Assisted **140+** students through problem-solving and crib sessions, facilitating clear understanding of concepts & improving academic support
- Facilitated smooth coordination between professors & students, communicating course requirements, student concerns and academic updates

EXTRACURRICULAR

Sports:

- Completed a **12-months** rigorous **National Sports Organization** (NSO) Cricket training camp underwent regular practice sessions
- **Champions** in the inaugural Mixed Chemical Engineering Department Cricket 2023 Tournament**,** securing the title in the first-ever edition

Arts:

- Built and grew **500+** subscribers & **160K+** views YouTube channel, Narcosed, demonstrating content creation & audience engagement
- Secured **#2** institute-wide in the Diesta 75-Hour Filmmaking Challenge contributing to video editing & post-production under sharp deadline

Technical:

- Participated in a **multi-objective optimization** competition, leveraging mathematical & statistical models to analyze experimental data
- Attended a **BioPharma workshop**, gaining insights into pharmaceutical industry workflows and the integration of AI/ML in drug discovery